28 July 2026

DIGESTIVE SYSTEM

Digestion ; The process of breaking down large food molecules into smaller molecules to obtain energy.

Digestive system ; A group of organs through which digestion takes place.

- Digestion takes place from enzymes.
- Digestion takes place in alimentary canal.
- Length of the alimantary canal - 30-33 feet approximately 10 meters.

alimantary canal

↓

Oral cavity → teeth - tongue saliva

↓

Pharynx

↓

Food pipe

↓

Stomach

↓

Small intestine → Duodenum, jejunum, illeum

↓

Large intestine → cecum, colon, rectum

29 July 2026

1. Digestion in mouth

1. Teeth

→ study of teeth → Odentology / Dentology

→ Teeths are made of dentin; Calcium Carbonate

Dentin → $CaCO_3$
→ Enamel

→ Outer white part / upper part of the teeth; dentin.

→ Hardest part of human body → Enamel



→ In gums.
↳ muscles + blood capilary.

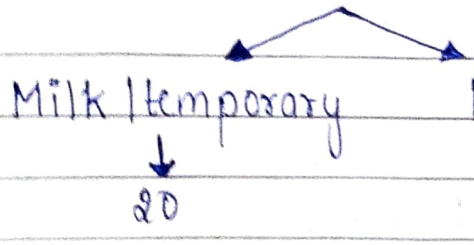
→ Organisms that grow teeth only once in their lifetime → Monophyodont.

→ Organisms that grow teeth twice in their lifetime → Diphyodont.

Monophyodont → Eg. → Difin walrus
Dolphin
Kangaroo
Molars in human.

Diphyodont → Maximum mammals.
Some reptiles

- Total teeth → 32



→ Based on function teeth are of 4 types →

1. Incissors
   ↳ Function → cutting

- More development in rats and squirrels.
- First teeth to appear in children.
- From the outer tusks of an elephant.

2. Canine
   ↳ Function → Piercing (भेदन) and tearing (चीरना)
   More development in carnivores.

3. Pre molar
   ↳ Function → Chew / masticate

4. Molar
   ↳ Function → Grinding & mixing

Dental formula in adults

$$\frac{I\ \ C\ \ P.M\ \ M}{2\ \ 1\ \ 2\ \ 3} \times 2$$

Dental formula in children

$$\frac{I\ \ C\ \ PM\ \ M}{2\ \ 1\ \ 0\ \ 2} \times 2$$

Note → 4 teeth emerge at the age of 22|24|26 year which are called Akkal Daddb / Wisdom teeth.

2. Tongue

→ It is a muscular organ
→ Not a single bone found in it.
→ But it is connected to the throat bone, the hyoid bone.

Function →
- Helps in speaking
- Detecting taste
- Helps in mixing food with saliva.

- Taste buds are found on the tongue to detect taste.

number → 2000 - 8000

Part - 4
Anterior (front part) → sweet taste
Posterior (Back part) → Bitter taste
Lateral (Side part) → Salty taste
Central (middle part) → Sour taste

3. Saliva

Secretion → From the salivary gland
Salivary glands → (i) Parotid salivary gland - below ear
→ Largest salivary gland.
→ Poison gland in snake.
(ii) Sublingual salivary gland - below tongue
(iii) Submandibular salivary gland - Below mandible (jaw bone).

- pH value → 6.8 - 7.2
- Nature → Basic
- Quantity → 1.5 - 2 litre / day.

- Enzyme → 1. Lysozyme → To destroy harmful bacteria that enter with food.

2. Salivary Amylase → To digest carbohydrates.

3. Ptyalin / Diastase
starch → maltose.

Note → Lysozyme bacteria are also present in tears and sweets.

→ Digestion start in mouth → carbohydrate
→ 70% carbohydrate digest in mouth.

→ 99% of saliva consists of water, and the rest contains enzymes.
→ The digested slimy food substance in the mouth - Bolus.
→ The bolus reaches the stomach passing through the pharynx and esophagus.
→ No digestion takes place in the pharynx and esophagus.

② Digestion in stomach
- Study of stomach - Gastrology
- Gland of stomach - Gastric gland
- Examination of stomach - Gastroscopy / Endoscopy
- Shape - J-shape

- Juice secreted from the stomach – Gastric Juice.

→ Gastric glands are found in the walls of the stomach in which different types of cells are formed.

like : Zymogen cell
Oxyntic cell
Mucus cell
Gastric cell

Gastric juice
pH value – 1.5 - 2
Nature – Acidic

Gastric juice
- Water (99%.)
- Substance in dissolved state (27%)
  - Organic / carbonic
    - Pepsin
    - Rennin
    - Lipase
    - Gelatinase
    - Urease
  - Inorganic
    - Bicarbonates
    - HCl, Na, K, Ca

31 July 2026 (Friday)

(i) Pepsin → To digest protein
Protein → Peptones and peptides.

) Lipase → To digest fat
Fat —Lipase→ Fatty acid + glycerol

ii) Rennim
Protein present —Rennin→ Paracaseinate
in milk

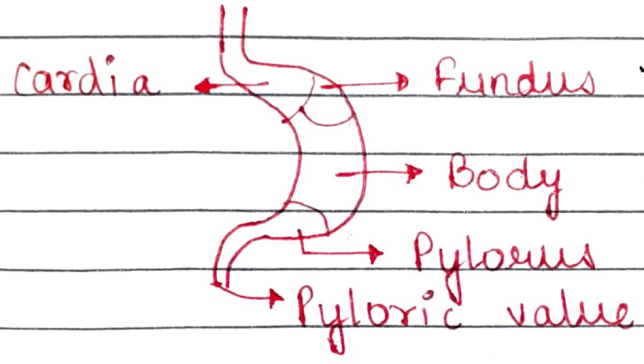
→ More active in children and calves.
→ Inactive in adult.
→ Industrial production of rennim in done in white biotechnology.
→ Used in the rasgulla and cheese industry.

(iv) HCl (Hydrochloric acid)
→ Convert inactive pepsinogen into active pepsin.
→ Makes the food medium acidic.
→ Destroyes unnecessary bacteria.

→ Desease caused by the excess of HCl - Acidity
→ Medicine used in the treatment of acidity - Antacid
→ Magnesium hydroxide - in making antacid.
→ The secretion of gastric juice increases as a reflex action as soon as food in seen
→ Digestion of fat and protein also beings in the stomach.
→ Food digested in the stomach - Chyme
→ The substance called myosin present in the gastric juice provides lubrication to the walls of the stomach and maintain the peristaltic movement of food

→ Capacity of stomach → generally → 75-100ml after consuming food 1-3 litre.

The stomach has 4 part
Fundus
Body
Cardia
Pylorus.



→ Food enters the small intestine when the phyloric value open.
→ Stomach connected the small intestine by Phyloric value

③ Digestion in small intestine
length → 6 - 6.5 m

→ There are 3 part of small intestine ;
ⓐ Deudenum → liver
ⓑ Jejunum → Pancreas
ⓒ Ilium → Gall Bladder.

Ⓐ Digestion in Deudenum
(i) liver
study of liver → Hipetology

Cell of liver → (i) Hepatic cell → function
→ formation of liver.

(ii) kuffer cell → function
function due to
of toxic elements.

→ Alcohol, medicine & poison are digest in liver.

1 August 2026 (Saturday)

Location → Right side of the abdominal cavity.
Weight → 1.3 - 1.5 kg.

- Divideds into 2 lobes
- Largest gland in the body
- Largest internal organ.

→ Also known as biological laboratory of the body

function → To produce bile juice.

→ Bile Juice
pH value → 7.7 - 7.8
Nature → slightly alkaline
Colour → Green
Quantity → 500 ml - 1 lit / day.

Bile Juice

- Water (98%)
- Other dissolved substance (2%)
  - Organic substance
    - Bile pigments
      - (i) Bilirubin
      - (ii) Biliverdin
    - Bile Salt
      - (i) Sodium Glycocholate
      - (ii) Sodium Taurocholate
    - Cholestrol
    - Fatty acids
    - Lecithin
    - Myosin
  - Inorganic substance
    - Na, K, Ca, Bicarbonates

- Bile juice is green due to biliverdin pigment.
- Bilirubin level increases during jaundice disease.

Function of Bile Juice

- Emulsification of fats
- Absorption of fat-soluable vitamins.
- Bile juice contain no enzymes so it is called juice without enzymes.

(ii) Gallbladder
- Function → To store bile juice
- Size → pear shape

(iii) Pancrease
→ It is a mixed gland
  - Endocrine → 95%
  - Exocrine → 5%.

→ Small cells are found inside it.
→ Langerhans discovered these cells.
→ These are called "Islets of Langerhans".

Pancreas
- Endocrine Gland
  - α cell → Glucagon hormone
    - Increase the sugar level in blood
    - Carbohydrate → Glucose
  - β cell → Insulin hormone
    - Decrease the sugar level in blood
    - Glucose → Glycogen
  - δ-cell → Somatostatin hormone
    - To control physical growth by controlling the level of G.H.
- Exocrine gland
  - f-cell → Pancreatic Juice

→ Insulin and glucagon hormone together regulates sugar level.

→ Disease caused by deficiency of Insulin
  → Diabetes.

- Symptoms (Diabetes) → Frequent Urination
  → Frequent hunger
  → Frequent thirst

- Diabetes day → 14th November

- Normal sugar level in blood
  → 60-90 mg / 100 ml → Befor eating
  → 90-120 mg / 100 ml → After eating

→ Low blood sugar level → Hypoglycernia
→ High blood sugar level → Hyperglycernia

→ Insulin
  └→ Protein formed by 51 Amino acids.
→ Discovered by → Fredric Senger
→ Zn element present in it.
→ Vaccine discovered → Best & Batting.

- Juice secreated from E. cell → Pancreatic Juice.
  └→ pH value → 7.8-8.5
  nature → Basic.

→ It is called as complete digestive juice.
→ 3 enzymes found in it.

(i) Amylase → Digestion of carbohydrate.

Carbohydrate —30%→ Glucose —100%→ Glycogen

(ii) Trypsin → Digestion of protein
Protein —10%→ Peptones —10%→ Amino acids

(iii) Lipase → Digestion of remaining 5% of fats.
Lipid / fats —5%→ fatty acids —100%→ glycerol.

Note → Maximum digestion takes place in the duodenum (small intestine).

→ Digested food in the duodenum - chyle

→ Jejunum → To transport digested food to ileum.

→ Ileum → Fingers like structures are found in the ileum which are called intestinal villi.

On these, the absorption of digested food takes place in the form of glycogen.

Note → Maximum absorption of food takes place in the ileum (small intestine).

→ Transport of energy obtained from digested food to various organs → Assimilation.

→ The ileum part of small intestine connected to the caecum part of large intestine by ileocecal valve.

→ Order of length → ileum > jejunum > Duodenum.

(4) Digestion in large intestine

- length → 1.5 - 2 metre
- 3 parts → (i) Ceacum
  (ii) colon
  (iii) Rectum

(i) Ceacum

- function → To deliver digested food into the colon.

→ The appendix is attached to this.
→ Digestion of cellulose takes place here

(ii) Colon

→ Water absorption takes place.
→ E-coloi bacteria found in it.
→ It obtain it own food from our undigest food.
→ It produce vitamin D,
E-coloi ⟷ (symbiotic relⁿ) ⟷ Human
waste food.

(iii) Rectum → Feces is stored in it.

It is connected in it anus through which feces is eliminated from the body.

Energy → 1 gm - carbohydrate → 4.0 calory
1 gm - protein → 4.5 calory
1 gm - lipid → 9.3 calory.